

Software Requirement Specification (SRS)
Smart AI-Based Urban Street Lighting Management System

CO2 Assessment Tool 1
Scenario-Based Requirement Analysis

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1. Introduction

1.1 Purpose

This Software Requirement Specification (SRS) document defines the functional and non-functional requirements, actors, use cases, data needs, and interface requirements for a Smart AI-Based Urban Street Lighting Management System. It serves as the foundation for design, development, testing, and validation of the proposed system for municipal corporations.

1.2 Document Conventions

Functional requirements are identified using the prefix "FR-" and use cases using the prefix "UC-", each followed by a two-digit sequence number. Priority levels (High/Medium/Low) indicate relative implementation urgency.

1.3 Intended Audience

- Municipal corporation IT and infrastructure teams
- Software architects and development team
- Quality assurance and testing engineers
- Project stakeholders and municipal decision-makers

1.4 Project Scope Overview

The system replaces fixed-schedule street lighting with an adaptive, AI- and IoT-driven platform that monitors light status, detects faults automatically, optimizes brightness in real time, predicts maintenance needs, and provides a centralized dashboard for municipal staff — improving public safety while reducing energy and maintenance costs.

1.5 References

- IEEE Std 830-1998 – Recommended Practice for Software Requirements Specifications
- Assigned Problem Statement: "Smart AI-Based Urban Street Lighting Management System" (CSA10 CO2 Assessment)

2. Problem Description

2.1 Business Domain

The system operates in the Smart City / Municipal Public Infrastructure domain, specifically urban street lighting operations managed by municipal corporations. It integrates IoT sensor networks, cloud computing, and AI-based analytics to modernize a traditionally manual, schedule-driven public utility service.

2.2 Existing Challenges

- Fixed lighting schedules cause unnecessary energy consumption during low-traffic hours.
- Malfunctioning streetlights are identified only through manual inspection or public complaints, causing delays.
- Lack of centralized visibility makes it difficult for administrators to track the health of city-wide lighting infrastructure.
- Reactive (rather than predictive) maintenance increases repair costs and downtime.
- Absence of data-driven reporting limits municipalities' ability to plan energy budgets or sustainability initiatives.

2.3 System Objectives

- Continuously monitor the operational status of every networked streetlight.
- Automatically detect faulty or malfunctioning streetlights.
- Use AI to optimize lighting intensity based on real-world conditions.
- Reduce overall electricity consumption without compromising public safety.
- Predict maintenance requirements before failures occur.
- Generate actionable energy usage and performance reports.
- Support broader smart city initiatives through open, integrable architecture.

3. Scope of the System

3.1 In-Scope

- Real-time monitoring of streetlight status via IoT sensors.
- AI-based automatic brightness optimization and fault detection.
- Predictive maintenance analytics and work-order management.
- Centralized web-based dashboard with map visualization.
- Automated alerting/notification for maintenance teams.
- Energy usage and cost-saving reporting.
- Role-based access control and a limited citizen-facing portal.

3.2 Out-of-Scope

- Physical installation or replacement of streetlight hardware (handled by field operations, not the software system).
- Billing and payment processing for municipal electricity contracts.
- Traffic signal control systems (a separate smart-city subsystem).

- Non-lighting municipal assets (e.g., water supply, waste management).

4. Stakeholders

The following individuals and groups have an interest in, or are affected by, the system:

Stakeholder	Interest / Role
Municipal Corporation	Owns and funds the system; responsible for city-wide energy and safety outcomes.
Administrators / Municipal IT Staff	Configure, monitor, and manage the system and its policies.
Field Maintenance Staff	Execute repairs based on system-generated alerts and work orders.
Citizens / Public	Beneficiaries of improved safety; may report issues via the public portal.
Energy Regulators / Auditors	Review energy consumption and sustainability reports.
Software Development Vendor	Designs, builds, and maintains the platform.

5. Functional Requirements

The following functional requirements define the specific behaviors and capabilities the system shall provide:

ID	Requirement	Description	Priority
FR-01	Streetlight Status Monitoring	The system shall continuously receive and display the ON/OFF status, brightness level, and power draw of every networked streetlight via IoT sensors.	High
FR-02	Automatic Fault Detection	The system shall automatically detect malfunctioning, flickering, or non-responsive streetlights and flag them as faults in real time.	High
FR-03	AI-Based Brightness Optimization	The system shall use an AI model to adjust streetlight brightness dynamically based on traffic density, pedestrian movement, ambient light, and weather conditions.	High
FR-04	Predictive Maintenance	The system shall analyze historical performance and sensor degradation data to predict components likely to fail and recommend preventive maintenance schedules.	High

ID	Requirement	Description	Priority
FR-05	Centralized Monitoring Dashboard	The system shall provide a web-based dashboard displaying a live map of all streetlights, their status, energy consumption, and active alerts.	High
FR-06	Fault Alert Notification	The system shall automatically generate and send alerts/notifications (SMS, email, or app push) to maintenance staff when a fault is detected.	High
FR-07	Maintenance Work-Order Management	The system shall allow administrators to create, assign, track, and close maintenance work orders linked to detected faults.	Medium
FR-08	Energy Usage Reporting	The system shall generate periodic (daily/weekly/monthly) energy consumption and cost-saving reports, exportable in PDF/Excel formats.	Medium
FR-09	User Authentication & Role-Based Access	The system shall authenticate users and restrict features based on roles (Administrator, Maintenance Staff, Viewer/Citizen).	High
FR-10	Configurable Lighting Rules	The system shall allow authorized administrators to configure brightness thresholds, scheduling rules, and zone-specific lighting policies.	Medium
FR-11	Historical Data Logging	The system shall log all sensor readings, faults, and maintenance actions with timestamps for audit and analytics purposes.	Medium
FR-12	Public/Citizen Portal	The system shall provide a limited public-facing view or complaint-reporting interface for citizens to report non-functional streetlights.	Low

6. Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints of the system:

Category	Requirement
Performance	The system shall process sensor data and update the dashboard within 3 seconds and support at least 50,000 concurrently connected streetlight nodes.

Category	Requirement
Security	The system shall encrypt all data in transit (TLS 1.2+) and at rest, and enforce role-based access control and secure device authentication for IoT nodes.
Reliability	The system shall maintain 99.5% uptime for the monitoring and alerting services and gracefully handle sensor/network failures without data loss.
Usability	The dashboard shall be intuitive for non-technical municipal staff, requiring no more than 2 hours of training, and shall support responsive access from desktop and mobile devices.
Scalability	The architecture shall support horizontal scaling to accommodate city-wide expansion from a pilot zone to the entire municipality without redesign.
Maintainability	The system shall use modular, well-documented microservices so that new AI models, sensors, or reporting modules can be integrated with minimal disruption.
Availability	Core monitoring and alerting services shall be available 24/7, with redundant cloud infrastructure and automatic failover.
Interoperability	The system shall expose open APIs to integrate with existing Smart City platforms, GIS systems, and municipal ERP/billing systems.

7. Actors and Use Cases

7.1 Actors

Actor	Type	Description
Municipal Administrator	Primary	Configures lighting rules, monitors the city-wide dashboard, generates reports, and manages user access.
Field Maintenance Staff	Primary	Receives fault alerts, executes repairs, and updates work-order status from a mobile interface.
Citizen / Public Portal User	Primary	Views public lighting status and reports non-functional streetlights via the public portal.
AI/Analytics Engine	Secondary (System Actor)	Processes sensor data to optimize brightness and predict maintenance needs; interacts autonomously with the system.

Actor	Type	Description
IoT Sensor & Smart Lighting Controller	Secondary (System Actor)	Streams real-time light intensity, status, and environmental data, and executes brightness/on-off commands.

7.2 List of Use Cases

Use Case ID	Use Case Name	Associated Actor(s)
UC-01	Monitor Streetlight Status	IoT Sensor & Smart Lighting Controller, Administrator
UC-02	Detect Faulty Streetlight	IoT Sensor & Smart Lighting Controller, System
UC-03	Optimize Lighting Intensity (AI)	AI/Analytics Engine
UC-04	Predict Maintenance Needs	AI/Analytics Engine
UC-05	Generate Energy Usage Report	Administrator
UC-06	View Centralized Dashboard	Administrator, Citizen
UC-07	Schedule Maintenance Work Order	Administrator, Field Maintenance Staff
UC-08	Send Fault Alert Notification	System, Field Maintenance Staff
UC-09	Authenticate User / Manage Access	Administrator, Field Maintenance Staff, Citizen
UC-10	Configure Lighting Rules & Thresholds	Administrator

7.3 Use Case Diagram

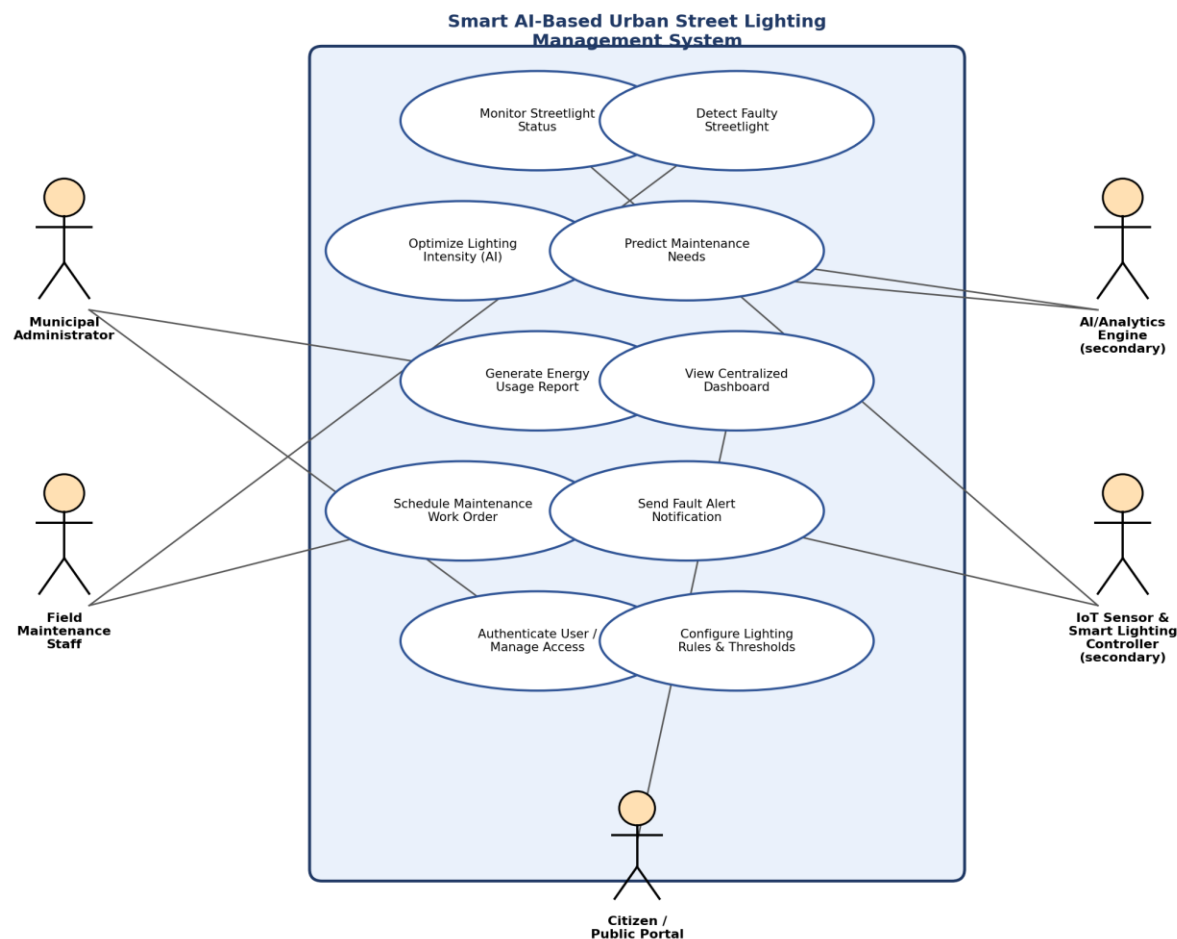


Figure 1: Use Case Diagram Smart AI-Based Urban Street Lighting Management System

7.4 Detailed Use Case Descriptions

UC-01: Monitor Streetlight Status

Use Case ID	UC-01
Use Case Name	Monitor Streetlight Status
Actors	IoT Sensor & Smart Lighting Controller (primary trigger), Municipal Administrator (viewer)
Preconditions	Streetlight sensors are installed, powered, and registered on the network.
Main Flow	1. IoT sensor periodically transmits status, brightness, and power data to the cloud platform. 2. System updates the streetlight's real-

	time status in the database. 3. Dashboard reflects the updated status on the map view.
Alternate / Exception Flow	If a sensor fails to transmit within the expected interval, the system marks the streetlight status as "Unknown" and logs a connectivity warning.
Postconditions	Streetlight status is current in the system and visible on the dashboard.

UC-02: Detect Faulty Streetlight

Use Case ID	UC-02
Use Case Name	Detect Faulty Streetlight
Actors	IoT Sensor & Smart Lighting Controller, System (automated)
Preconditions	Streetlight is registered and reporting status data.
Main Flow	1. System continuously analyzes incoming sensor data against expected operating parameters. 2. System identifies anomalies (e.g., zero output while scheduled ON, abnormal power draw, flickering). 3. System flags the streetlight as "Faulty" and creates a fault record. 4. System triggers the "Send Fault Alert Notification" use case.
Alternate / Exception Flow	If sensor data is ambiguous or intermittent, the system schedules a re-check before confirming the fault to avoid false positives.
Postconditions	A fault record is created and linked to the streetlight asset; maintenance staff are notified.

UC-03: Optimize Lighting Intensity (AI)

Use Case ID	UC-03
Use Case Name	Optimize Lighting Intensity (AI)
Actors	AI/Analytics Engine
Preconditions	Real-time traffic, pedestrian, and weather data feeds are available.
Main Flow	1. AI engine ingests traffic density, pedestrian movement, ambient light, and weather data. 2. AI model computes the optimal brightness level per zone/streetlight. 3. System sends brightness adjustment commands to the smart lighting controller. 4. Controller applies the new brightness level.

Alternate / Exception Flow	If AI confidence is low or data feeds are unavailable, the system falls back to the pre-configured default schedule.
Postconditions	Streetlight brightness is adjusted to the optimal level for current conditions.

UC-04: Predict Maintenance Needs

Use Case ID	UC-04
Use Case Name	Predict Maintenance Needs
Actors	AI/Analytics Engine
Preconditions	Historical sensor and fault-history data is available for the asset.
Main Flow	1. AI engine analyzes historical performance trends and degradation patterns. 2. System calculates a failure-risk score for each streetlight component. 3. System generates a predictive maintenance recommendation and adds it to the maintenance queue.
Alternate / Exception Flow	If insufficient historical data exists for a newly installed asset, the system defers prediction until a minimum data threshold is met.
Postconditions	A predictive maintenance recommendation is available for administrator review and work-order scheduling.

UC-05: Generate Energy Usage Report

Use Case ID	UC-05
Use Case Name	Generate Energy Usage Report
Actors	Municipal Administrator
Preconditions	Administrator is authenticated with reporting privileges.
Main Flow	1. Administrator selects a reporting period and zone/scope. 2. System aggregates energy consumption and cost-saving data. 3. System generates a report and offers export in PDF/Excel format.
Alternate / Exception Flow	If no data exists for the selected period, the system displays an appropriate empty-state message.
Postconditions	A report is generated, displayed, and optionally downloaded.

8. Data Requirements

The system shall capture, store, and manage the following categories of data:

Data Entity	Key Attributes
Streetlight Asset	Asset ID, location (GPS), zone, installation date, hardware model, current status
Sensor Reading	Timestamp, light intensity, power draw, ambient light level, connectivity status
Fault Record	Fault ID, asset ID, detection timestamp, fault type, severity, resolution status
Maintenance Work Order	Work order ID, assigned staff, priority, scheduled date, completion date, notes
Energy Report	Report period, zone, total consumption (kWh), cost, savings vs. baseline
User Account	User ID, role, credentials (hashed), contact details, access permissions
AI Model Output	Prediction ID, asset ID, risk score, recommended brightness, model version, confidence score

Data shall be retained for a minimum of 3 years for audit and trend-analysis purposes, and archived per municipal data retention policy thereafter.

9. External Interface Requirements

9.1 User Interfaces

- Responsive web dashboard for administrators (desktop and tablet).
- Mobile application/interface for field maintenance staff (Android/iOS).
- Simplified public web portal for citizens to view status and report issues.

9.2 Hardware Interfaces

- Interfaces with IoT light sensors, smart LED drivers, and lighting controllers over standard protocols (e.g., Zigbee, LoRaWAN, or NB-IoT).
- Supports gateway devices that aggregate data from streetlight clusters before cloud transmission.

9.3 Software Interfaces

- Cloud platform services (compute, storage, and managed database).
- AI/ML model-serving service for optimization and prediction algorithms.
- Integration APIs with existing Smart City / GIS / municipal ERP platforms.

9.4 Communication Interfaces

- Secure HTTPS/TLS REST APIs for dashboard and mobile app communication.
- MQTT or CoAP protocol for lightweight, low-power IoT sensor communication.
- SMS/Email/Push notification gateway integration for alerting.

10. System Features

Feature	Description	Priority
Real-Time Monitoring	Live map-based view of all streetlight statuses across the city.	High
AI Brightness Optimization	Automatic, condition-based brightness adjustment to save energy.	High
Automated Fault Detection & Alerts	Detects malfunctions and instantly notifies maintenance teams.	High
Predictive Maintenance	Forecasts component failures to enable proactive servicing.	High
Energy & Cost Reporting	Periodic reports on consumption, savings, and sustainability metrics.	Medium
Role-Based Access Control	Differentiated access for administrators, staff, and citizens.	High
Work Order Management	End-to-end tracking of maintenance tasks from creation to closure.	Medium
Public Complaint Portal	Citizen-facing channel to report non-functional streetlights.	Low

11. Assumptions and Constraints

11.1 Assumptions

- All streetlights to be monitored will be retrofitted or newly installed with compatible IoT sensors and smart controllers.

- Reliable network/cellular coverage is available across the municipal area for IoT data transmission.
- Municipal staff will receive basic training to operate the administrator dashboard.
- Historical energy consumption data is available or can be reasonably estimated for baseline comparison.
- The municipal corporation will provide continued cloud infrastructure funding for system operation.

11.2 Constraints

Type	Constraint
Technical	System must integrate with existing legacy streetlight infrastructure where full hardware replacement is not feasible.
Operational	Field maintenance staff availability and response times may vary by zone, affecting fault-resolution SLAs.
Economic	Initial IoT sensor deployment and cloud infrastructure costs are constrained by the municipal budget.
Legal	The system must comply with data privacy regulations for any citizen-submitted data collected via the public portal.
Time	Phase 1 (pilot zone deployment) must be completed within the municipal fiscal-year planning cycle.

12. Requirement Validation

12.1 Completeness Check

Requirements were cross-checked against each stakeholder group and each stated project objective. All eight objectives from the problem statement (continuous monitoring, fault detection, AI optimization, energy reduction, predictive maintenance, reporting, public safety, and smart-city support) map to at least one functional requirement and one use case, confirming stakeholder needs are addressed.

12.2 Consistency Check

Functional requirements were reviewed for conflicts (e.g., ensuring AI-driven brightness optimization, FR-03, does not contradict the minimum-safety brightness constraints implied by public safety objectives). No contradictory requirements were identified; configurable thresholds (FR-10) were added specifically to let administrators enforce safety minimums that the AI model must respect.

12.3 Ambiguity, Redundancy & Missing Requirement Check

- Ambiguity: The original problem statement's phrase "optimize brightness automatically" was clarified into a specific requirement (FR-03) with defined input factors (traffic, pedestrian movement, weather) to remove interpretive ambiguity.
- Redundancy: Fault detection (FR-02) and alerting (FR-06) were kept as separate requirements to avoid conflating detection logic with notification logic, improving testability.
- Missing Requirements Identified: Role-based access control (FR-09) and audit data logging (FR-11) were added, as the original scenario implied but did not explicitly state authentication and traceability needs.

12.4 Suggested Improvements

- Define explicit Service Level Agreements (SLAs) for fault-resolution time per fault severity level.
- Add a requirement for offline/degraded-mode operation when cloud connectivity to a sensor cluster is temporarily lost.
- Incorporate a feedback loop where citizen-reported complaints (UC via public portal) are automatically cross-validated against AI-detected faults to improve model accuracy.
- Specify measurable energy-reduction targets (e.g., "reduce consumption by X% within 12 months") to make performance NFRs independently verifiable.